

Detecting Response Styles in Slider Scales

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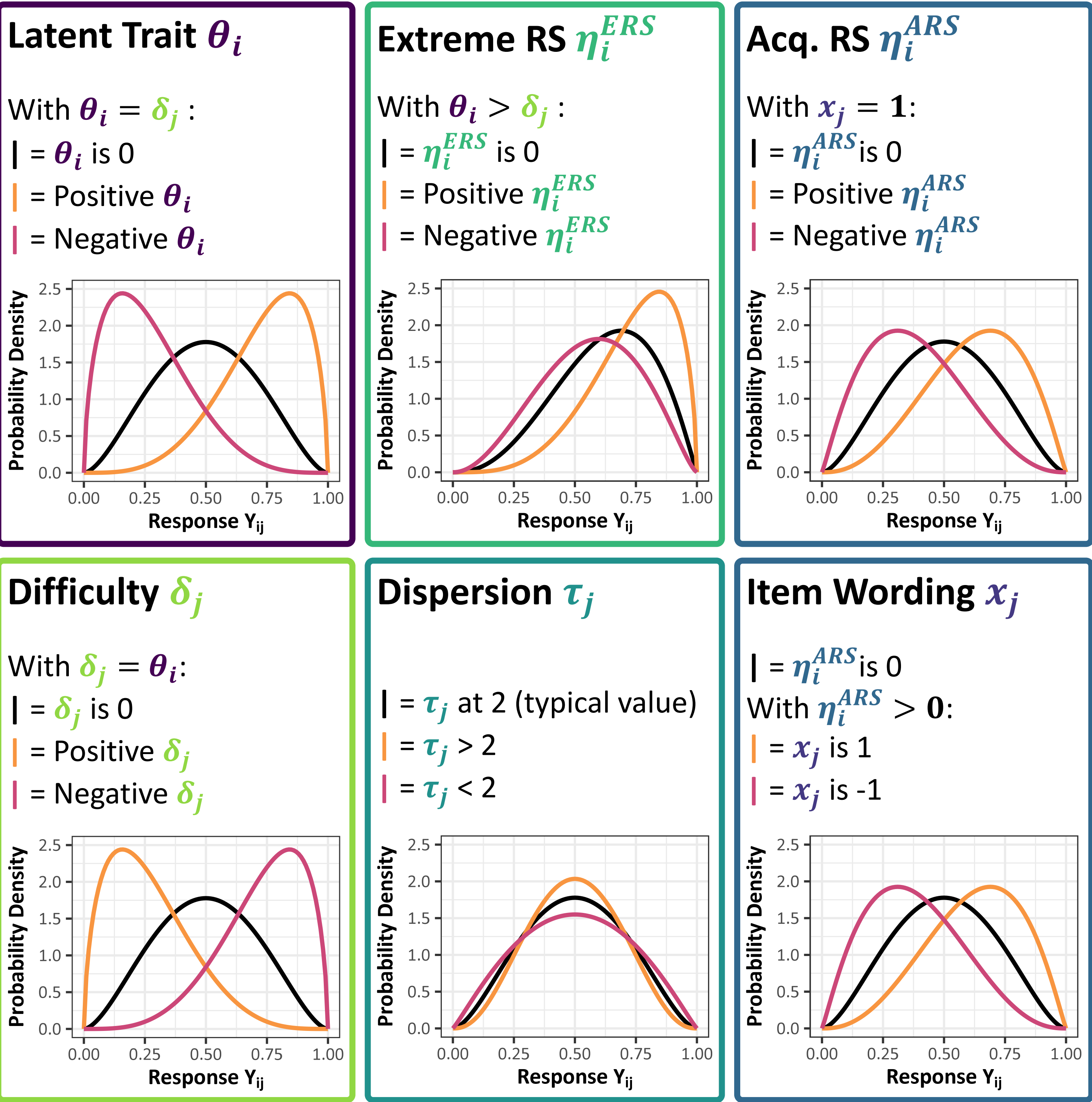
Introduction: Beta Item Response Model (BIRM)

- Original model by Noel & Dauvier (2007)
- Extended to detect Extreme Response Style (ERS) and Acquiescent Response Style (ARS)

BIRM With Response Styles (BIRM-RS)

- i : Person, j : Item

$$m_{ij} = \exp\left(\frac{\exp(\eta_i^{ERS}) \cdot (\theta_i - \delta_j + x_j \cdot \eta_i^{ARS}) + \tau_j}{2}\right),$$
$$n_{ij} = \exp\left(\frac{-\exp(\eta_i^{ERS}) \cdot (\theta_i - \delta_j + x_j \cdot \eta_i^{ARS}) + \tau_j}{2}\right),$$
$$Y_{ij} = \text{Beta}(m_{ij}, n_{ij})$$



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We developed a model to detect response styles in slider scales.

Try out the Shiny-App!

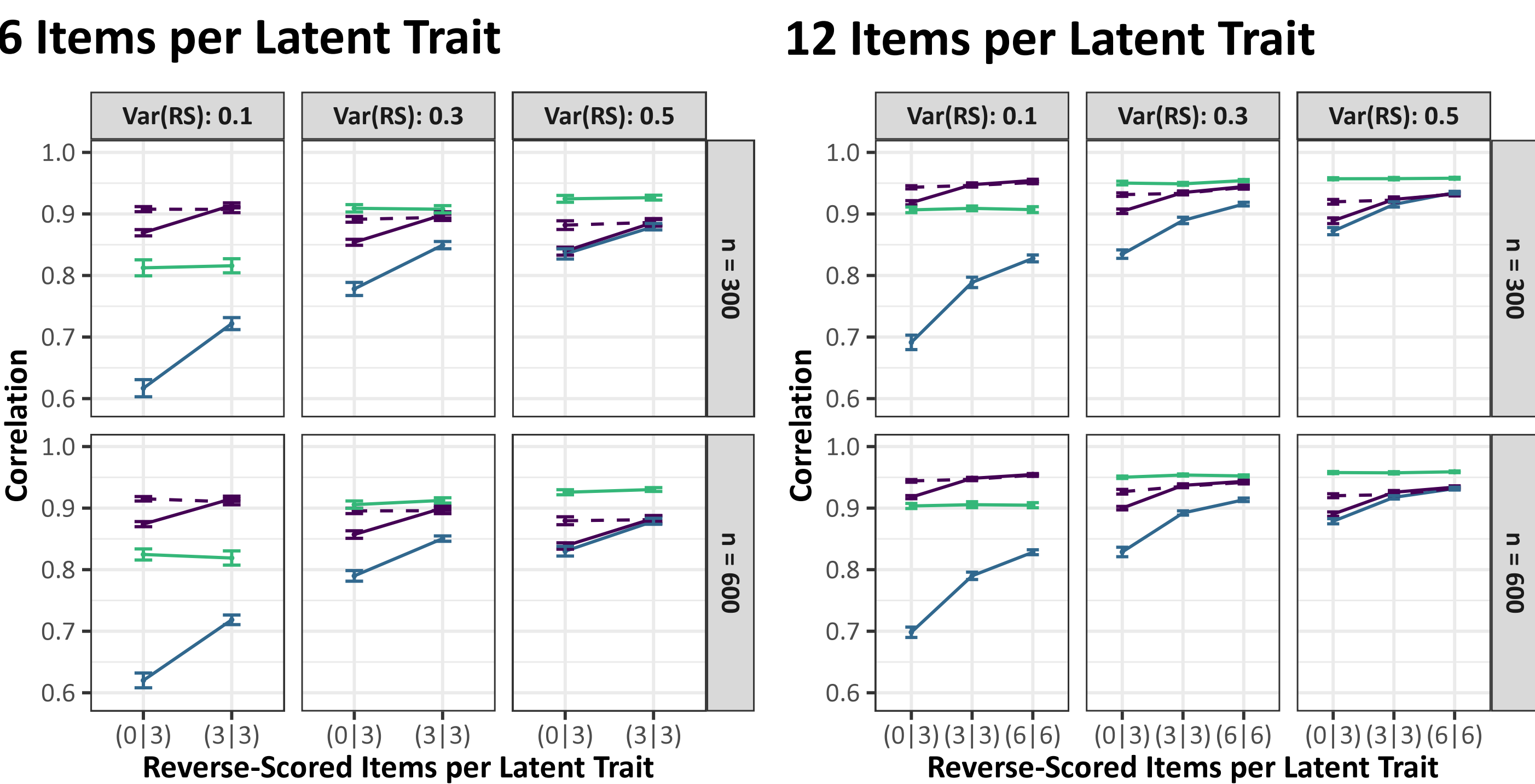
Simulation Study Design

- Goal: Parameter Recovery
- MCMC sampling implemented in Stan
- Two latent traits modeled

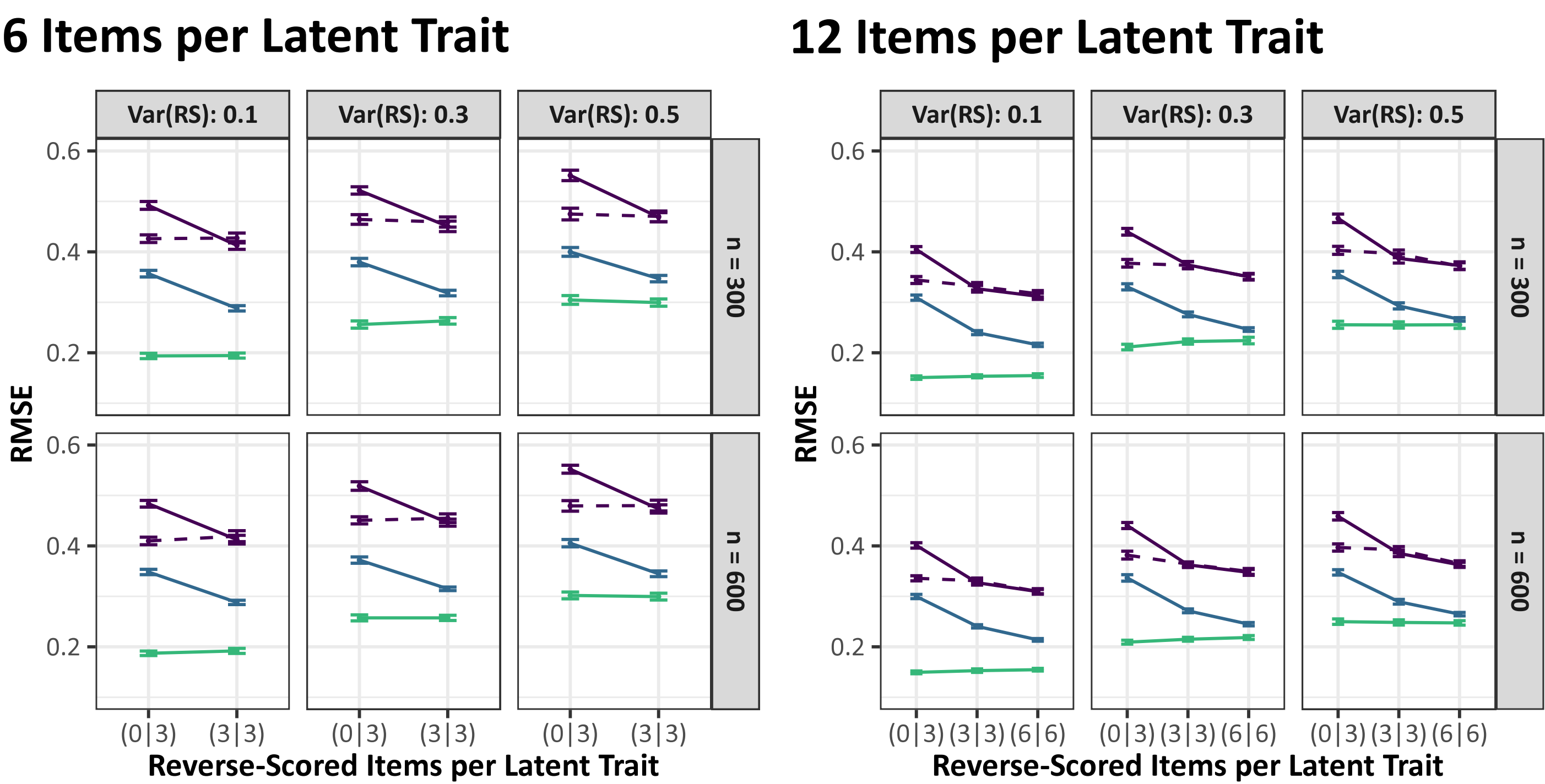
Results

- $|$ = First latent trait
- $|$ = Second latent trait
- $|$ = Extreme RS
- $|$ = Acquiescent RS

Correlations Between True and Estimated Values



RMSEs of True and Estimated Values



Discussion

- Model successfully recovers parameters given sufficient items and reverse-scored items
- Empirical illustration planned

Reference

Noel, Yvonnick, & Dauvier, Bruno (2007). A Beta Item Response Model for Continuous Bounded Responses. *Applied Psychological Measurement*, 31(1), 47–73. <https://doi.org/10.1177/0146621605287691>.